Test Paper

Instructions:

- Answer all questions.
- Write your code in the provided space.
- Use 3.x syntax.

Section A: Multiple Choice Questions (1 point each)

1. What is the output of the following code?
   ```
   print(type([]) == list)
   ```
   a) True
   b) False
   c) Error
   d) None
2. Which of the following is a mutable data type in ?
   a) Tuple
   b) String
   c) List
   d) Int
3. What will be the result of the expression `3 * 'abc'`?
   a) abcabcabc
   b) 3abc
   c) None
   d) Error
4. Which keyword is used to define a function in ?
   a) func
   b) define
   c) def
   d) function
5. What will the following code output?

   ```
   x = [1, 2, 3]
   print(x[1:3])
   ```
   a) [1, 2]
   b) [2, 3]
   c) [1, 2, 3]
   d) Error
6. Which of the following statements creates a set?
   a) s = {1, 2, 3}
   b) s = [1, 2, 3]
   c) s = (1, 2, 3)
   d) s = 1, 2, 3
7. How do you start a comment in ?
   a) //
   b) /*
   c) #
   d) <!--
8. Which of the following is used to handle exceptions in ?
   a) try...except
   b) if...else

c) catch
d) handle

9. What will be the output of the following code?

```
x = {1, 2, 2, 3}
print(len(x))
```

a) 2
b) 3
c) 4
d) Error

10. What does the len() function do?
a) Returns the last element of a list
b) Returns the number of elements in an iterable
c) Calculates the length of a string
d) Both b and c

Section B: Short Answer Questions (2 points each)

11. Explain the difference between append() and extend() methods in lists.
12. What is a lambda function in ? Provide an example.
13. Describe how you can read a file in .
14. What is the purpose of the with statement in file handling?
15. Explain the difference between deepcopy and shallow copy.

Section C: Coding Problems (5 points each)

16. Write a function named fizz_buzz that takes an integer n as input and prints the numbers from 1 to n. For multiples of three, print "Fizz" instead of the number, for multiples of five print "Buzz", and for multiples of both three and five print "FizzBuzz".

```
def fizz_buzz(n):
# Write code in Jupyter Notebook
```

17. Given a list of numbers, write a function get_even_numbers that returns a new list containing only the even numbers from the input list.

```
def get_even_numbers(numbers):
# Write code in Jupyter Notebook
```

18. Write a program to read a file and count the number of words in it. Assume the file name is provided as input.

```
def count_words_in_file(file_name):
# Write code in Jupyter Notebook
```

19. Write a function to check if a given string is a palindrome.

```
def is_palindrome(s):
# Write code in Jupyter Notebook
```

20. Create a function that returns the Fibonacci sequence up to n numbers.

```
def fibonacci(n):
# Write code in Jupyter Notebook
```

21. Write a function that takes a list of integers and returns the maximum and minimum values.

```
def max_min(numbers):
# Write code in Jupyter Notebook
```

22. Create a function to find the factorial of a number using recursion.

```
def factorial(n):
# Write code in Jupyter Notebook
```

23. Write a function that accepts a string and counts the frequency of each character.

```
def char_frequency(s):
# Write code in Jupyter Notebook
```

24. Write a program that sorts a list of dictionaries based on a specified key.

```
def sort_dicts(list_of_dicts, key):
# Write code in Jupyter Notebook
```

Section D: Debugging (3 points each)

25. Identify and correct the error in the following code snippet:

```
def multiply_list(my_list):
  result = 1
  for i in my_list:
    result * i
  return result
```

26. The following function is intended to return the factorial of a number, but it contains an error. Identify and fix it.

```
def factorial(n):
  if n == 0:
    return 1
  else:
    return n * factorial(n)
```

27. Identify the error in the following code and provide a corrected version:

```
def add_numbers(x, y):
  return x + y

result = add_numbers(5)
```

28. There is a logical error in this function. Identify it and suggest a correction:

```
def find_max(numbers):
  max_num = numbers[0]
  for num in numbers:
    if num > max_num:
      max_num = num
  return max_num
```

29. Find and fix the error in the following code:

```
my_list = [1, 2, 3]
print(my_list[3])
```

30. This code is meant to return a list of squares of numbers but has an issue. Identify it:

```
def square_numbers(nums):
    return [n**2 for n in nums]
```

Total Marks: 100